The Vanishing of Diffeomorphism Hamiltonians

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$$\text{Diff(M)} \Rightarrow \text{Noether II} \Rightarrow J\xi \approx dQ\xi \Rightarrow H\xi = \int \partial\Sigma (Q\xi - \xi \cdot B) \Rightarrow \partial\Sigma = \emptyset \Rightarrow H\xi = 0$$

This chain is one of the most profound results in the geometry of gauge theories.

1. Diff(M) ⇒ Noether II

Because diffeomorphisms form an **infinite-dimensional gauge group** — symmetries parametrised by arbitrary vector fields ξ ∈ Vect(M) — Noether's *second* theorem applies, not the first. Where Noether I gives conserved currents for global symmetries, Noether II gives **differential identities among the Euler–Lagrange equations** for local/gauge symmetries.

Concretely, for any ξ ∈ Vect(M), invariance of the action under x ↦ x + ξ yields an off-shell identity:

$$Ea \cdot \mathcal{L}_\xi \varphi a \equiv \nabla\mu \, J\mu\xi \qquad \textit{(off-shell)}$$

This is not a conservation law — it holds by virtue of the gauge structure alone, before imposing equations of motion.

2. Noether II ⇒ Jξ ≈ dQξ

The Noether current associated to ξ takes the form:

$$J\xi = \Theta(\varphi, \mathcal{L}_\xi \varphi) - \xi \cdot L$$

where Θ is the symplectic potential from the boundary term δL = E·δφ + dΘ. The key consequence of Noether II is that **on-shell** (≈), this current is not merely closed but actually **exact**:

$$J\xi \approx dQ\xi$$

Qξ is the **Noether charge (n−2)-form**. Its existence follows from the off-shell identity: the gauge structure forces Jξ to be trivial in cohomology on-shell. In GR with ξ a Killing vector, Qξ = −(1/16πG) εμν∇μξν — the Komar form.

3. Jξ ≈ dQξ ⇒ Hξ = ∫∂Σ(Qξ − ξ·B)

In the covariant phase space formalism, the Hamiltonian generating the flow of ξ is defined via:

$$\delta H\xi = \int_\Sigma \omega(\varphi, \delta\varphi, \mathcal{L}_\xi \varphi)$$

where ω = δΘ is the pre-symplectic form. Using Jξ ≈ dQξ and Stokes' theorem, this bulk integral localises to a boundary integral:

$$H\xi = \int \partial\Sigma \, (Q\xi - \xi \cdot B)$$

The **B-term** arises from ambiguities in Θ (defined up to dB) and ensures integrability on phase space. This is the **Iyer–Wald formula**. The diffeomorphism Hamiltonian is pure boundary — there is no bulk piece. This is the gravitational avatar of the fact that the stress-energy of gravity is a pseudotensor.

4. ∂Σ = ∅ ⇒ Hξ = 0

If the spatial slice Σ is a closed manifold without boundary — as in a compact universe, or when boundaries are pushed to infinity with sufficiently fast field decay — then ∂Σ = ∅ and Stokes' theorem gives nothing:

$$H\xi = 0$$

5. What This Actually Means

This vanishing is not a failure of the formalism. It is a deep structural feature of diffeomorphism-invariant theories:

Interpretation	Content
Constraints	Hξ = 0 is the diffeomorphism constraint (and the Hamiltonian constraint for ξ = n̂, the normal). The Hamiltonian is the constraint.
No local energy	Gravity has no gauge-invariant local energy density. Energy is only defined relationally, at boundaries.
ADM / BMS charges	When ∂Σ ≠ ∅ (asymptotic boundaries), Hξ recovers ADM mass, angular momentum, and BMS charges — all as boundary integrals.

6. The Problem of Time

The equation Hξ = 0 in the bulk is precisely the content of the **Wheeler–DeWitt equation** in canonical quantum gravity: the physical Hilbert space consists of states annihilated by all constraints. Time, energy, and dynamics as ordinarily conceived must be reconstructed from correlations — the **problem of time**.

The boundary is not a technicality. It is the physics.